

Milestone 2 Final Report

AI-Powered Soil Analytics System for Nutrient Assessment and Intelligent Crop Advisory

Scope: CNN Soil Image Analysis, Structured Nutrient-Deficiency Modeling, Explainable AI, and Hybrid Soil Analysis

1. Milestone Overview

Milestone 2 develops the machine-learning and explainability components of the soil analytics system. The work combines a CNN-based soil-image classifier with structured machine-learning models for N, P, and K deficiency estimation, followed by SHAP, Grad-CAM, and a hybrid assessment pipeline.

2. Image Data and CNN Analysis

The cleaned image dataset contains 1,139 images across seven soil classes: Alluvial_Soil, Arid_Soil, Black_Soil, Laterite_Soil, Mountain_Soil, Red_Soil, and Yellow_Soil. Images were converted to RGB and resized to 224 × 224 pixels. The dataset was split into 793 training images, 168 validation images, and 178 test images.

EfficientNetB0 with ImageNet pretrained weights was selected as the CNN backbone. The initial transfer-learning stage froze the pretrained base and trained a seven-class classification head. The model was subsequently fine-tuned by unfreezing selected upper layers with a lower learning rate.

Model	Test Accuracy
Initial EfficientNetB0	90.45%
Fine-tuned EfficientNetB0	92.13%

The fine-tuned EfficientNetB0 was selected as the final CNN model and saved as `efficientnet_soil_classifier_finetuned.keras`. Evaluation included accuracy, precision, recall, F1-score, confusion matrix analysis, and incorrect-prediction inspection.

3. Structured Soil Machine Learning

The structured dataset contains 2,200 observations and 23 columns. The original dataset label represents crop recommendation rather than nutrient-deficiency status. Because ground-truth N, P, and K deficiency labels were not available, the project derived dataset-relative proxy labels using the 25th percentile of each nutrient.

The resulting thresholds were N = 21, P = 28, and K = 20. Values below the corresponding threshold were assigned the Deficient class, while values at or above the threshold were assigned Not Deficient. These proxy labels are not laboratory-confirmed agronomic deficiency measurements.

For each nutrient, the nutrient used to construct its own target was excluded from the corresponding feature matrix to prevent direct target leakage. An 80/20 stratified train-test split was used. Gradient Boosting classifiers were trained as baselines and tuned using 5-fold cross-validation with F1 as the tuning metric.

Nutrient	Selected Model	Accuracy	Precision	Recall	F1
N	Tuned Gradient Boosting	0.7500	0.4815	0.3645	0.4149
P	Baseline Gradient Boosting	0.9273	0.8333	0.8796	0.8559
K	Tuned Gradient Boosting	0.8091	0.5857	0.4271	0.4940

The final N, P, and K models were saved as separate pickle files in the Models directory. The baseline P model was retained because it produced a slightly higher held-out test F1 than its tuned counterpart; for K, tuning improved recall

and F1 and the tuned model was retained.

4. SHAP Explainability

SHAP (SHapley Additive exPlanations) was applied to the final Gradient Boosting models using TreeExplainer. Global summary and feature-importance plots were generated for N, P, and K deficiency models. Local waterfall explanations were also generated for individual predictions.

The SHAP analysis identifies features that influence the trained model's output. For example, the global analyses highlighted variables such as humidity, potassium, rainfall, phosphorus, nitrogen, temperature, pH, and other structured features depending on the nutrient model.

Interpretation limitation: SHAP explains model behavior; it does not establish causality. A positive SHAP contribution means that a feature pushed the model output toward the modeled positive class for the relevant prediction, not that the feature caused nutrient deficiency.

5. Grad-CAM Explainability

Grad-CAM was applied to the final fine-tuned EfficientNetB0. The selected late feature layer produced a $7 \times 7 \times 1280$ activation representation. Gradients of the selected class score were used to weight feature maps and generate a normalized heatmap that was resized and overlaid on the original 224×224 image.

Both a correct and an incorrect prediction were examined. The correct example was classified as Alluvial_Soil with approximately 73% confidence. The incorrect example had actual class Alluvial_Soil and predicted class Laterite_Soil with approximately 47.8% confidence.

Interpretation limitation: Grad-CAM highlights image regions associated with the model prediction, but it does not prove that the highlighted regions are causal or that the CNN has learned scientifically validated soil properties.

6. Hybrid Soil Analysis

The hybrid system integrates the two model branches at the assessment level. The image branch predicts soil type and image confidence, while the structured branch predicts N, P, and K deficiency status and confidence. Their probabilities are not directly averaged because the branches predict different targets.

A project-defined nutrient-status score is calculated from the three modeled nutrient statuses: 100 for 3/3 nutrients not deficient, 66.67 for 2/3, 33.33 for 1/3, and 0 for 0/3. This score is an internal project indicator and is not a scientifically validated soil-health index.

Sample	Soil Type	N	P	K	Score
0	Alluvial_Soil	Not Def.	Not Def.	Not Def.	100.00
500	Alluvial_Soil	Deficient	Not Def.	Deficient	33.33
1000	Alluvial_Soil	Not Def.	Not Def.	Not Def.	100.00
1500	Alluvial_Soil	Not Def.	Not Def.	Not Def.	100.00
2000	Alluvial_Soil	Not Def.	Not Def.	Not Def.	100.00

7. Limitations

- N/P/K deficiency labels are dataset-relative proxy labels rather than laboratory-confirmed ground truth.
- The hybrid dataset does not provide verified pairings between individual soil images and individual structured soil-test records.
- The project-defined soil-health score should not be interpreted as a validated agronomic soil-health index.
- Model confidence values are reported probabilities and should not automatically be interpreted as calibrated real-world certainty.

- The image classifier is a soil-type classification model and is not a replacement for laboratory soil testing.

8. Final Project Structure

Data/ → Raw, Processed, Metadata

Models/ → final CNN and N/P/K Gradient Boosting models

Notebooks/ → 01 through 08 project notebooks

Outputs/ → CNN, SHAP, GradCAM, and Hybrid artifacts

Reports/ → dataset documentation and this Milestone 2 report

9. Conclusion

Milestone 2 establishes the project's core machine-learning pipeline: image-based soil classification, structured N/P/K deficiency estimation, model explainability through SHAP and Grad-CAM, and an integrated hybrid assessment. The final implementation and supporting outputs are organized for subsequent project development and review, with the documented methodological limitations retained as part of the project's scope.